Explainable NLP for Caption Similarity and Hashtag Recommendation

Paper

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Abstract

On TikTok, captions and hashtags shape how content gets discovered, but it is still unclear whether posts with similar captions also attract similar levels of engagement. This paper asks a specific question: do semantically similar captions—measured through transformer-based sentence embeddings—cluster into neighborhoods where engagement metrics are more consistent than in randomly assembled groups? And if so, can that regularity be used to recommend hashtags in an explainable way?

To answer this, 22,647 TikTok posts were encoded using Sentence-BERT (`all-MiniLM-L6-v2`), producing 384-dimensional vectors. K-nearest-neighbor neighborhoods were built at $K \in \{5, 10, 20\}$, and the within-group variance of four log-transformed engagement metrics (likes, comments, views, shares) was compared against null distributions from 1,000 random groups per neighborhood size. All twelve Mann-Whitney U tests came back significant ($p \approx 0.0$), with variance reductions between 10.5% and 13.0%. Four classification baselines converged at 59–60% accuracy, which confirms that captions alone are not enough to predict engagement—but the structural pattern is real. A retrieval-based hashtag recommender built on these neighborhoods produces topically relevant suggestions, each one traceable to actual posts in the corpus.

The main takeaway is that semantic similarity does not predict *how much* engagement a post will get, but it does predict *how consistently* similar posts perform. That structural regularity turns out to be enough for a practical, transparent recommendation system.

1 Introduction

On short-form platforms like TikTok, captions and hashtags are the main textual handles for content discovery. Transformer-based language models have made it possible to measure semantic similarity between these short texts with real precision, but an important question has been mostly overlooked: do captions that *mean* similar things also *perform* similarly? Most prior work treats engagement as a prediction problem—feed in the text, predict the numbers [2, 9]. That framing makes sense for many applications, but it is not the only way to think about the relationship between text and engagement.

The research question here is different: *Does semantic similarity between captions, measured through transformer embeddings, correspond to similarity in engagement behavior? And can that correspondence support an explainable hashtag recommendation system?* The key shift is from prediction to structure. Instead of asking whether a caption can predict a view count, the question is whether captions that land close together in embedding space also end up with more homogeneous engagement distributions than random groupings [12]. This is a subtler question, and it has received almost no attention in the literature despite having clear practical implications for recommendation systems built around transparency.

The approach works as follows. Captions are encoded into dense vectors using a transformer-based sentence embedding model. These vectors make it possible to group semantically similar captions into neighborhoods. The analysis then checks whether those neighborhoods have lower within-group engagement variance than groups assembled at random—focusing on structural alignment rather than direct prediction. For hashtag recommendation, the hashtags used by posts in each neighborhood are aggregated and ranked by frequency and engagement performance. Because every recommendation points back to real historical posts, the system is inherently transparent. Section 2 covers the relevant literature; Section 3 lays out the data and methods; Section 4 presents the findings; and Section 5 wraps up with contributions, limitations, and what could come next.

2 Literature Review

The question driving this project sits at the intersection of three research areas: how to measure semantic similarity in short texts, how engagement on social media relates to linguistic properties, and what it means for a recommendation system to be explainable. Each of these areas is reviewed below, with a focus on the specific gaps that the present work is designed to address.

2.1 Semantic Similarity in Short Texts: From Meaning to Vector Space

Before anything else, the project needs a way to measure how similar two captions are in terms of meaning. In modern NLP, the standard approach is to map sentences into a high-dimensional vector space, where geometric closeness stands in for semantic closeness. The most common measure is cosine similarity:

$$\text{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|} \tag{1}$$

What this formula does is compare the angle between two vectors rather than their magnitude. Two captions that point in roughly the same direction in embedding space are considered semantically similar, regardless of how long or detailed they are. For TikTok, that means a caption like "Just hit a new personal record at the gym" and one like "Smashed my fitness goals today" should end up with a high cosine similarity, even though they share almost no words.

The real breakthrough in making this work came with BERT [4], which introduced contextualized token-level representations. BERT captures nuances like word sense and syntactic structure in ways that previous models could not. But BERT was not designed for comparing whole sentences efficiently. Reimers and Gurevych [14] point out that using BERT as a cross-encoder for sentence comparison creates "a massive computing overhead"—searching through just 10,000 sentences could take around 65 hours. Their solution, Sentence-BERT (SBERT), uses siamese and triplet network architectures to produce fixed-size sentence embeddings that can be compared directly with cosine similarity [14]. This is what makes it feasible to compute pairwise similarities across tens of thousands of captions in reasonable time.

Building on this, Gao et al. [7] introduced SimCSE, a contrastive learning framework that improves embedding quality even further. The interesting part is how simple the approach is: unsupervised SimCSE works "with only standard dropout used as noise" [7]. But the authors also note that the method is fragile—"removing [dropout] leads to a representation collapse," meaning the model produces identical embeddings for everything. This is a useful warning: the quality of the embedding space is not guaranteed, and if the geometry breaks down, any downstream analysis based on nearest neighbors becomes unreliable.

SimCSE reports strong gains on standard semantic textual similarity benchmarks, measured by Spearman correlation against human annotations [7]. That is encouraging, but there is an important caveat. These benchmarks evaluate whether embeddings agree with human judgments about *meaning* similarity. They say nothing about whether meaning similarity tracks *behavioral*

similarity—such as engagement patterns. That gap between semantic fidelity and behavioral correlation is exactly what this project is trying to test.

2.2 Retrieval Foundations: Sparse, Dense, and Scalable Similarity Search

Once captions have been embedded, the practical task becomes retrieval: given a new caption, find the most similar ones in the corpus. The retrieval literature offers two main paradigms, and the choice between them is not straightforward.

On the lexical side, BM25 has been a workhorse for decades. Robertson and Zaragoza [15] describe it as "one of the most successful text-retrieval algorithms," and its staying power is notable. BM25 works by matching exact terms, weighted by frequency and document length. It is fast, well-understood, and surprisingly hard to beat in many settings. Dense retrieval methods like SBERT take a different approach—they encode queries and documents into the same vector space and measure cosine similarity, which allows them to capture meaning even when the words differ.

The BEIR benchmark [16] was created specifically because too many retrieval models were being evaluated in "homogeneous and narrow settings." The main finding is sobering: BM25 remains "a robust baseline," and neural models sometimes outperform it but "at considerable computational costs." For this project, the takeaway is that both lexical and semantic similarity deserve to be evaluated. If only one representation captures the engagement structure, that itself is an informative finding.

On the scalability side, FAISS [10] makes it practical to search over large vector collections. Johnson et al. describe billion-scale similarity search as fundamental for handling "complex data such as images or videos." The dataset here is much smaller (around 22,000 posts), but FAISS still provides a clean, efficient implementation of exact cosine search over normalized vectors. Once captions become fixed-size vectors, the retrieval problem is purely geometric.

2.3 Engagement Modeling in Social Media

The second part of the research question is about engagement. On social media, engagement is usually measured through likes, comments, shares, and views. These metrics are driven by a complex mix of factors: the text itself, but also audio choices, visual quality, follower count, posting time, and algorithmic amplification [6]. Text is only one piece of a much larger puzzle.

Most existing research frames engagement as a prediction problem. Lamprinidis, Hardt, and Hovy [11] study news headline popularity and find that it depends on "topical, structural, and lexical factors." They show that even "a simple Logistic Regression model over character n-grams" can pick up meaningful engagement signals from text. This suggests that language does carry some predictive power, even if it is limited.

But there is a distinction that matters a lot for the current project: predicting engagement from text is not the same as showing that similar texts have similar engagement distributions [9].

Prediction is a mapping problem—input text, output number. The structural question is different: it asks whether posts that are close in embedding space also tend to have similar engagement profiles. A classifier might hit 60% accuracy while still missing the fact that semantically coherent groups are internally more consistent than random groups. Conversely, strong within-group consistency does not mean engagement is predictable from text.

Engagement metrics also differ in what they capture. Likes can be relatively passive, while comments and shares often involve deeper cognitive effort [3]. This is important because semantic similarity might correlate more strongly with some engagement dimensions than others. The evaluation in this project tests all four metrics independently for exactly that reason.

2.4 Hashtag Recommendation in Micro-Video Contexts

The second half of the research question asks whether the structural relationship between similarity and engagement can be turned into something practical: hashtag recommendations. Hashtag recommendation has a long history, especially on Twitter and in news streams [5]. Recent work describes hashtags as "very large and dynamic" with a "variable relevance lifecycle" [8]—they trend, plateau, and fade, often within days. This makes recommendation under temporal drift particularly difficult.

In micro-video contexts, the challenge intensifies. Wang et al. [17] note that micro-videos "are usually associated with hashtags" and that good hashtags "can increase the probability of their micro-videos to be discovered." But they also highlight two key obstacles: long-tail distributions, where most hashtags are used only once or twice, and multimodal complexity, since micro-videos combine visual, acoustic, and textual signals. Current state-of-the-art systems typically fuse all these modalities using graph neural networks [18], achieving high performance but sacrificing interpretability.

A text-only approach cannot compete with these multimodal systems on raw accuracy. What it can do is serve as a hypothesis-driven investigation: how far can caption text alone take the analysis? If semantic neighborhoods turn out to have meaningful engagement structure, then hashtag recommendations drawn from those neighborhoods have a principled justification. That link between structural validity and recommendation quality is what connects this section to the final pillar of the literature review.

2.5 Explainable Recommendation and Example-Based Justification

The last theoretical piece is explainability. What makes a recommendation system explainable? Zhang and Chen [21] define the goal as producing "not only high-quality recommendations but also understandable explanations." They draw a clear line between two approaches: post-hoc explanations, which are generated after the fact to rationalize an opaque model's output, and intrinsic explanations, which "directly come from an explainable model."

A retrieval-based system fits naturally into the second category. When a hashtag is recom-

mended because it appeared frequently among the K nearest semantic neighbors, the explanation is built into the mechanism itself: here are the similar posts, here are the hashtags they used, here is how they performed. No hidden latent factors, no black-box reasoning. The justification is example-based and fully auditable [12].

Stepping back, the literature paints a clear picture. Transformer embeddings produce reliable semantic representations of short texts. Linguistic features have measurable effects on engagement. Retrieval-based systems offer a natural path toward explainability. But one question has not been asked directly: does proximity in embedding space correspond to proximity in engagement behavior? That is the gap this project addresses, and the next section describes the methodology used to test it.

3 Methodology

This section covers the dataset, how text was represented, how semantic neighborhoods were built, and how the statistical tests and recommendation mechanism work. The overall goal is to test whether captions that are semantically close also show more consistent engagement patterns than random groups.

3.1 Dataset and Preprocessing

The data comes from a custom TikTok scraper, stored in a PostgreSQL database on Supabase.[1] The database has four tables in a normalized schema: `videos` (caption and creation timestamp), `video_snapshots` (likes, comments, plays, shares, and scrape timestamp), `video_hashtags` (a bridge table linking videos to hashtag IDs), and `hashtags` (the actual tag strings). A SQL CTE joins these tables and picks the most recent engagement snapshot for each video.

The raw dataset contains 26,623 TikTok posts. After removing 3,976 posts (14.9%) whose captions had fewer than four non-hashtag characters—essentially posts that were nothing but hashtag strings with no real caption—22,647 posts remained.

One design decision turned out to be critical: how hashtags are handled. The process has two phases, and the order matters. First, hashtags are *extracted* from both the bridge table and the inline caption text. The bridge table alone covered only 883 posts (3.9%), which is far too sparse for any recommendation system to work. After also pulling hashtags from captions, coverage jumped to 100%—an average of 5.58 tags per post across 36,866 unique tags. Second, once hashtags have been extracted, they are *removed* from the caption text before encoding. This is important because if hashtag strings stayed in the text, the embedding model would pick up on hashtag-string similarity instead of actual topical meaning. A post tagged `#fitness` would look similar to another post tagged `#fitness` even if the captions were about completely different things. The hashtag is the label, not the feature.

Engagement variables (likes, comments, views, shares) were log-transformed to handle the extreme skew typical of social media data:

$$y' = \log(1 + y) \tag{2}$$

In concrete terms, a post with 500,000 views gets mapped to $y' = \log(1 + 500{,}000) \approx 13.12$, while a post with 1,000 views becomes $y' \approx 6.91$. The "+1" inside the log prevents issues with posts that have zero engagement on some metric. Binary labels were created by splitting on the median of log-transformed views (median = 13.06), producing 11,324 high-engagement and 11,323 low-engagement posts—a perfectly balanced 50/50 split that gives classifiers a clear baseline to beat.

[1] `https://github.com/PredictiveSocialMedia/Tik-Tok-Recommendation-System`

3.2 Text Representations

Two different text representations were used, each capturing something different about the captions.

The first is TF-IDF, a classic bag-of-words approach that weights terms by how frequent they are in a document versus how common they are across the whole corpus. Formally:

$$\text{tf-idf}(t, d) = (1 + \log \text{tf}(t, d)) \times \log \frac{N}{\text{df}(t)} \tag{3}$$

Here, $\text{tf}(t, d)$ counts how often term t appears in document d, N is the total number of documents, and $\text{df}(t)$ is how many documents contain t. The sublinear formulation—using $1 + \log \text{tf}$ rather than raw counts—dampens the effect of terms that repeat many times in a single caption. The resulting matrix has dimensions $22{,}647 \times 10{,}000$ (posts by features), includes bigrams, and is very sparse (0.16% density). TF-IDF was fitted on demojized text, meaning emoji characters were first converted to descriptive tokens like `flexed_biceps`, so that emojis could be treated as regular vocabulary items.

The second representation is Sentence-BERT (SBERT), using the `all-MiniLM-L6-v2` model. This maps each caption to a fixed-size 384-dimensional vector:

$$\mathbf{e}_i = \text{SBERT}(c_i) \in \mathbb{R}^{384}, \quad \|\mathbf{e}_i\| = 1 \tag{4}$$

The key difference from TF-IDF is that SBERT captures *meaning* rather than just word overlap. Two captions can share zero words but still end up with nearly identical embeddings if they express similar ideas. All vectors are L2-normalized (unit length), which means that the dot product between any two vectors directly equals their cosine similarity. For SBERT, raw Unicode emojis were preserved in the input text, since the model handles them natively. UMAP was used to project the 384-dimensional space down to two dimensions for visualization purposes.

3.3 Semantic Neighborhoods and Variance Analysis

The heart of the analysis is the construction of semantic neighborhoods and the comparison of their engagement properties to random baselines. For each of the 22,647 posts, a neighborhood is the set of K closest posts in the SBERT embedding space:

$$N_K(q) = \{c_1, c_2, \ldots, c_K\} \quad \text{where} \quad c_j = \underset{c \in C \setminus \{q\}}{\arg\max}\ \text{sim}(\mathbf{e}_q, \mathbf{e}_c) \tag{5}$$

In plain terms, for each post q, the K posts whose embeddings have the highest cosine similarity to q's embedding are selected as its neighbors. Neighborhoods were computed for $K \in \{5, 10, 20\}$ using the FAISS library for fast exact cosine search over normalized vectors.

The comparison against random baselines works as follows. For each post's semantic

neighborhood, the within-group variance of each engagement metric is calculated. Then 1,000 random groups of the same size K are drawn from the corpus, and their variances are computed to form a null distribution. The question is whether the semantic neighborhoods consistently land on the low end of this distribution. Variance reduction quantifies this:

$$\Delta\sigma^2 = \frac{\bar{\sigma}^2_{\text{random}} - \bar{\sigma}^2_{\text{semantic}}}{\bar{\sigma}^2_{\text{random}}} \times 100\% \tag{6}$$

A positive value means that semantic neighborhoods have lower engagement variance than random groups—in other words, posts that mean similar things also perform more consistently.

Two statistical tests formalize this comparison. The one-sided Mann-Whitney U test checks whether the distribution of semantic neighborhood variances is stochastically smaller than the distribution of random group variances. This is a non-parametric test, meaning it does not assume normality—appropriate here because engagement distributions on social media are notoriously heavy-tailed. Levene's test adds a second layer by checking whether the two distributions differ in spread (dispersion), which helps distinguish between a shift in the average variance and a genuine difference in how the variances are distributed.

The hashtag recommendation component follows directly from the neighborhood structure. Given a new caption q, the candidate hashtag set is everything used by the neighbors:

$$H(q) = \bigcup_{c_i \in N_K(q)} H(c_i) \tag{7}$$

These candidates are ranked by two criteria: how often they appear among the K neighbors (frequency), and the average log-engagement of posts that use them. Frequency captures topical relevance—a hashtag used by many similar posts is probably a good match. Average engagement provides a performance signal. Because every recommendation points to specific real posts, the system is inherently explainable: a user can inspect *why* a particular hashtag was suggested and *which* posts it came from.

3.4 Classification Baselines

Four classifiers were trained to establish how much engagement-relevant signal actually exists in caption text. The data was split 80/20 with stratification (18,117 training; 4,530 test). The models are: Logistic Regression on SBERT embeddings, Multinomial Naïve Bayes on TF-IDF, and KNN at $k \in \{5, 10\}$ on SBERT.

The purpose of these classifiers is not to build the best possible engagement predictor. That would require multimodal features—audio, video, follower data—which are far outside the scope of a text-only analysis. The point is to measure where the ceiling is. If all four models, using two different feature representations, converge at roughly the same accuracy, that convergence is evidence that the bottleneck is in the data itself: captions simply do not contain enough

information to predict engagement well. And that interpretation has a direct consequence for the structural analysis. If the variance reduction finding were just an artifact of class separation—if high-engagement posts clustered together in embedding space—then the classifiers should be able to exploit that. The fact that they cannot makes the structural finding more interesting, not less.

4 Results

This section walks through four sets of results: the structure of the embedding space, how well classifiers can predict engagement from text, whether semantic neighborhoods actually have lower engagement variance than random groups, and how the hashtag recommendation system performs in practice.

4.1 Embedding Space and Feature Representations

The TF-IDF matrix came out to 22,647 × 10,000 with 0.16% density—very sparse, as expected for short texts with large vocabularies. SBERT produced 384-dimensional vectors for each caption. Figure 1 shows the UMAP projection of these embeddings down to two dimensions. The pattern is telling: a dense central cluster with scattered outliers, which is typical for a short-text corpus covering many different topics. What stands out is that high- and low-engagement posts are thoroughly mixed throughout the space. There is no obvious visual boundary between the two classes, which already suggests that engagement level is not something the embedding space can cleanly separate.

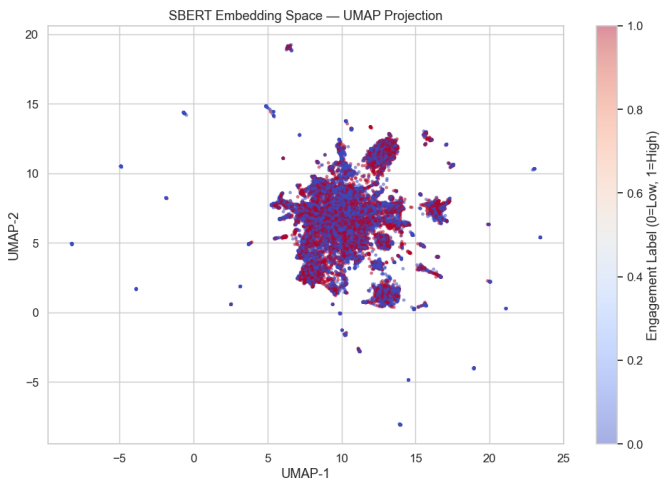

Figure 1: UMAP projection of SBERT caption embeddings. High-engagement (red) and low-engagement (blue) posts are intermixed, confirming that the embedding space does not separate posts by engagement level.

4.2 Classification Baselines: The Information Ceiling

Table 1 shows the results for all four classifiers. The headline number is the accuracy: 59–60% across the board, with macro F1 scores between 0.58 and 0.59. Given that the baseline (random guessing with balanced classes) is 50%, the models are doing something—but not much.

Table 1: Per-class classification results (test set, n = 4,530).

		Low engagement			**High engagement**		
Model	**Features**	P	R	F1	P	R	F1
Logistic Regression	SBERT	0.60	0.59	0.59	0.59	0.60	0.60
Multinomial Naïve Bayes	TF-IDF	0.66	0.41	0.50	0.57	0.79	0.66
KNN (k = 5)	SBERT	0.58	0.63	0.61	0.60	0.54	0.57
KNN (k = 10)	SBERT	0.57	0.73	0.64	0.62	0.44	0.51

The per-class numbers are more interesting than the overall accuracy. Naïve Bayes is heavily biased toward predicting high engagement—it catches 79% of high-engagement posts but only 41% of low-engagement ones. This likely reflects vocabulary patterns: popular posts tend to use certain words and phrases that NB picks up on, but it over-predicts as a result. KNN at k = 10 has the opposite bias, leaning toward low engagement. Logistic Regression is the most balanced of the four. Figure 2 makes these patterns easy to see.

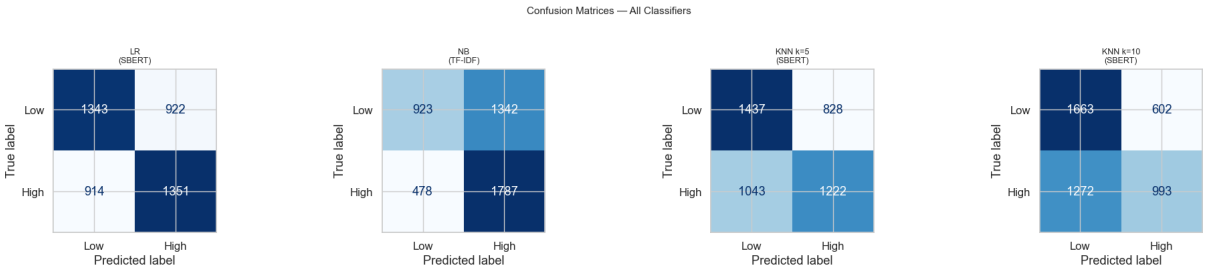

Figure 2: Confusion matrices for all four classifiers (test set, n = 4,530). Class 0 = low engagement; Class 1 = high engagement.

What matters most is the convergence. Both the dense representation (SBERT) and the sparse one (TF-IDF) hit the same wall at around 60%. If the richer semantic features of SBERT offered a genuine predictive advantage over keyword frequencies, the accuracy gap between the two should be noticeable. It is not. This points to an information ceiling: caption text, no matter how it is encoded, does not contain enough signal to reliably predict engagement on TikTok. Audio, visuals, follower networks, posting time, algorithmic boosting—all of these are likely doing most of the work, and none of them are present in the text. This ceiling is actually useful for the next analysis: it rules out the possibility that the variance reduction finding is just a byproduct of class separation in the embedding space.

4.3 Variance Reduction: The Core Finding

This is the central result. For all 22,647 posts, semantic neighborhoods were built at K = 5, 10, 20, and the within-group variance of each engagement metric was compared against 1,000 random

groups of the same size. Table 2 shows the outcome.

Table 2: Variance reduction (%) of semantic vs. random neighborhoods. All twelve Mann-Whitney U tests: $p \approx 0.0$.

K	likes_log	comments_log	views_log	shares_log
5	13.04%	12.72%	12.32%	12.46%
10	12.30%	12.74%	11.36%	11.98%
20	11.34%	11.27%	10.46%	10.53%

Every single value is positive, ranging from 10.46% to 13.04%. Every one of the twelve Mann-Whitney U tests returned $p \approx 0.0$. The finding is consistent across all four engagement metrics and all three neighborhood sizes.

The pattern across K values is also informative. Variance reduction decreases as K grows: roughly 12.6% at $K = 5$, dropping to about 10.9% at $K = 20$. This makes intuitive sense. At $K = 5$, only the five most similar posts are included—these share tight topical and stylistic overlap, and their audiences likely overlap as well. At $K = 20$, the boundary expands to include more distant posts, diluting the coherence. This monotonic decay is exactly what a genuine semantic–engagement relationship should produce: the signal is strongest when the neighborhood is tightest.

Levene's test added a further layer. At $K = 5$, the variance differences mainly reflect a shift in average variance ($p > 0.05$), meaning semantic neighborhoods are centered at a lower variance level but their variance distributions have a similar shape to the random ones. At $K = 10$ and $K = 20$, the dispersion itself becomes significantly different ($p < 0.05$). This suggests that at larger scales, semantic neighborhoods are not just shifted lower—they have a genuinely different distributional shape, with tighter concentration.

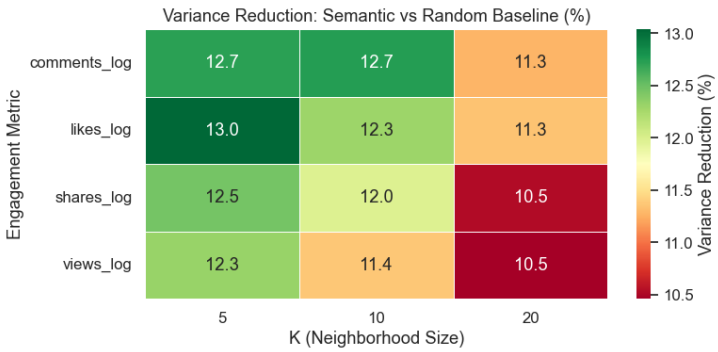

Figure 3: Heatmap of variance reduction (%) across engagement metrics and neighborhood sizes. The monotonic decrease from $K = 5$ to $K = 20$ reflects the dilution of semantic coherence as neighborhoods grow.

4.4 Hashtag Coverage and Recommendation

On the practical side, the two-phase hashtag extraction made a dramatic difference. Starting from 3.9% coverage with just the bridge table, the combined approach reached 100%—5.58 hashtags per post across 36,866 unique tags. Without this fix, building any kind of recommendation system would have been impossible for the vast majority of posts.

Table 3 shows what the recommender produces for three test captions, one each from fitness, finance, and food topics.

Table 3: Top hashtag recommendations for three unseen captions. Freq. = occurrences among K nearest neighbors.

Query caption	**Hashtag**	**Freq.**	**Avg. log-eng.**
"Just hit a new personal record at the gym! Hard work always pays off"	#fitness	7	14.2
	#gymtok	4	13.8
	#gym	3	13.5
"Crypto market update — here is my honest take on what is happening right now"	#stockmarket	6	13.1
	#cryptocurrency	4	12.9
	#crypto	3	12.7
"Healthy high-protein meal prep for the week, my go-to lunch routine"	#highprotein	3	13.4
	#mealprepideas	2	12.8
	#highproteinmeals	2	12.6

The results are intuitive. Fitness captions get fitness hashtags; crypto captions get finance hashtags; meal prep captions get food hashtags. None of this is surprising, but that is partly the point: the recommendations are sensible because the underlying neighborhoods are semantically coherent, and that coherence is statistically grounded in the variance reduction analysis. What makes this system different from a simple keyword matcher is that it operates in semantic space. A caption about "crushing it at the gym" would still get fitness hashtags even if it never uses the word "fitness," because SBERT encodes the meaning, not just the words.

5 Conclusion

The question behind this project was whether TikTok captions that are semantically similar also tend to attract similar engagement patterns. Across 22,647 posts, twelve independent Mann-Whitney U tests, four engagement metrics, and three neighborhood sizes, the answer is consistently yes. Variance reductions ranged from 10.5% to 13.0%, all with $p \approx 0.0$. At the same time, classification accuracy topped out at 59–60%, confirming that caption text alone cannot reliably predict *how much* engagement a post will get. The structural finding is about consistency, not magnitude: similar captions perform more *similarly*, even though none of the models can tell you whether a particular post will go viral.

Three contributions come out of this work. The first is the structural finding itself—the demonstration that semantic proximity in embedding space corresponds to engagement proximity on a short-form platform. This has not been shown before, and it holds even when classifiers fail to separate the engagement classes cleanly. The second is the hashtag recommendation system built on top of that finding. Every recommendation is grounded in real posts from the corpus and ranked by observable criteria (frequency and engagement), making the system transparent by construction. The third is methodological: the two-phase hashtag extraction that boosted coverage from 3.9% to 100%, and the variance-reduction-against-random-baseline evaluation protocol with 1,000 simulations per configuration.

There are clear limitations. The 59–60% accuracy ceiling underlines that engagement on TikTok is driven overwhelmingly by non-textual factors—audio, visuals, follower count, posting time, algorithmic amplification—and none of these appear in the data. The SBERT model (`all-MiniLM-L6-v2`) was trained mainly on English, so it likely underperforms on TikTok's substantial multilingual and slang-heavy content. The variance reduction itself varies with K, from about 13% at $K = 5$ to about 10.5% at $K = 20$, which means practitioners need to choose K based on how tight they want the neighborhoods. The dataset is a static snapshot, and adding new posts requires rebuilding the entire index. And dropping 14.9% of posts for having minimal captions introduces a bias toward posts that actually have something to say.

Looking ahead, the most impactful extension would be incorporating multimodal signals—audio embeddings, thumbnail features, video-level representations—to push beyond the text-only ceiling. Multilingual embedding models would broaden the system's reach. Incremental indexing through FAISS IVF partitions would make it possible to update the index without starting from scratch. And the strongest validation would come from A/B testing: deploying the recommended hashtags on real posts and measuring whether they actually influence engagement outcomes. The current work is a proof-of-concept, but the structural foundation it establishes—that meaning and performance are linked in a measurable, exploitable way—opens a clear path toward a practical tool for content creators.

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